Chandramauli Chakraborty

Curriculum Vitae

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Education

2024–present **PhD in Statistics**, *University of Chicago*
Advisor: Prof. Cong Ma.
Research focus: **RLHF, NLHF, and multi-turn reinforcement learning**. Broader interests include LLM post-training, AI agent alignment, and multimodal LLMs.

2022–2024 **Master of Statistics (M.Stat)**, *Indian Statistical Institute, Kolkata, India*
- First Division with Distinction.

2019–2022 **Bachelor of Statistics (B.Stat)**, *Indian Statistical Institute, Kolkata, India*
- First Division with Distinction.

2017–2019 **Higher Secondary Examination (Class 12)**, *Bhavan's Gangabux Kanoria Vidyamandir, Kolkata, West Bengal*
93.8% overall; 95% in Mathematics, Physics, and Chemistry.

2007–2017 **Secondary Examination (Class 10)**, *Don Bosco H.S. School, Silchar, Assam*
CGPA 10; A1 grade (above 95%) in Mathematics and Science.

Academic Achievements

2024 Received the **Neubauer Fellowship**, University of Chicago.

2024 Received an invitation for the **Sabyasachi Roy Memorial Gold Medal**.

2023 Received an **ACM India–IARCS travel grant**.

2022 Received an invitation for the **D. Basu Gold Medal Award**.

2019–2024 Received semester prizes and stipends from the Indian Statistical Institute, Kolkata.

2019 Qualified the **IISER Aptitude Test** and received an admission offer.

2019 Achieved **All India Rank 86** in the Banaras Hindu University Entrance Examination.

2019 Qualified **JEE Main and JEE Advanced**, ranking in the **top 2%**.

2019 Scored **790/800** in SAT Mathematics.

2018 Qualified the written examination of **Kishore Vaigyanik Protsahan Yojana (KVPY)**.

2017 Qualified the state-level **National Talent Search Examination (NTSE)**.

2016 **Regional Mathematics Olympiad (RMO)** awardee; appeared for the **Indian National Mathematics Olympiad (INMO)**.

Summer Projects and Research Experience

Jun–Aug 2023 Studied **generative modeling and theoretical aspects of rectified flow** with **Prof. Larry Wasserman** and **Prof. Arun Kumar Kuchibhotla**.

Jun–Aug 2022 Participated in the NIH-funded **Big Data Summer Institute (BDSI)** in the Department of Biostatistics, University of Michigan. Worked in the Data Mining group with Prof. Johann Gagnon-Bartsch, using **Python and spatial epidemiology** to forecast county-level deficits in reported STI cases associated with reduced surveillance testing during the pandemic. Training covered machine learning, Bayesian statistics, deep learning, NLP, missing data, model selection, and unsupervised learning.

Jun–Aug 2021 Worked on **Selecting Balls from Urns** in the *Polymath Jr. REU program* at Williams College, partly supported by the NSF, with Prof. Enrique Treviño. Investigated urn-sampling models and proved conjectures posed in *To Replace or Not to Replace—That is the Question*.

Jan 2021–Jan 2022 Worked on **COVID-19 modeling and public policy** with Prof. Gopal K. Basak, studying trade-offs between economic outcomes and lockdown duration.

Jan–Jun 2021 Studied **clustering and unsupervised learning** with Dr. Swagatam Das, including hard and soft clustering, Bregman and f-divergences, DP-means, spectral clustering, and sparse subspace clustering.

Academic Projects and Short Research Experiences

Spring 2022 Developed a **painting genre/style classifier** using hand-engineered visual features and clustering, with methods for analyzing paintings outside the genres represented in training. Completed for the Statistics Comprehensive course under Prof. Saurabh Ghosh and Prof. Bimal Kumar Roy.

Spring 2022 Studied *Triangle-free Graphs Have Large Independence Numbers* for the Random Graphs course under Prof. Mridul Nandi.

Fall 2021 Analyzed **Consumer Expenditure Survey** data from the National Sample Survey, adjusting variables of interest to compare consumption patterns across rural and urban populations, sex, and age groups.

Fall 2021 Analyzed the National Statistical Office's **Time Use Survey**, estimating policy-relevant indicators across rural and urban populations and by sex.

Spring 2021 Analyzed real-world **time series data** for Statistical Methods IV under Prof. Smarajit Bose.

Fall 2020 Analyzed *Fisher's Iris data* for Statistical Methods III, taught by Prof. Probal Chaudhuri.

Summer 2020 Completed a group project on the proof of *Weyl's Equidistribution Theorem* for Analysis II, taught by Prof. Pradipta Bandyopadhyay.

Jun 2020 Independently studied **backtracking algorithms** and implemented a **Sudoku Solver**.

Fall 2019 Analyzed *Galton's height data* for Statistical Methods I, taught by Prof. Probal Chaudhuri.

2017 Built a basic **Google Dino Game bot** in Python.

2016 Independently investigated long-division methods for computing nth roots.

Publications and Preprints

$\star$ denotes joint authorship.

- **Chakraborty, C.** and Ma, C. *The Statistical Benefits of Multiple Responses for Learning From Demonstrations*. Preprint; submitted to ICLR 2027.
- **Chakraborty, C.**$\star$, Lee, J.$\star$, and Ma, C. *Generative Collapse Does Not Imply Discriminative Collapse*. Submitted to ICLR 2027.
- **Chakraborty, C.**$\star$, Paul, S.$\star$, Chakraborty, S., and Das, S. *Automatic Convex Clustering of High-dimensional Data with Ordered Weighted ℓ_1 Regularization*. Manuscript completed.
- **Chakraborty, C.**$\star$, Paul, S.$\star$, Chakraborty, S., and Das, S. *Clustering RNA Single Cell Sequences: A Lasso-weighted k-means approach with Ordered Weighted ℓ_1 Regularization*. Manuscript completed.
- Burden, J., **Chakraborty, C.**, Fang, Q., Lin, L., Malibari, N., Matoush, S., Milbank, I., Parekh, Z., Prado, M., Ren, R., Rong, Q., Sánchez Garza, M., Sun, E., Treviño, E., and Yamada, D. *Selecting Balls From Urns With Partial Replacement Rules*. **PUMP Journal of Undergraduate Research**, 2024. Authors listed alphabetically.
- **Chakraborty, C.**$\star$, Paul, S.$\star$, Chakraborty, S., and Das, S. *Clustering High-dimensional Data with Ordered Weighted ℓ_1 Regularization*. **26th International Conference on Artificial Intelligence and Statistics (AISTATS)**, 2023.
- Basak, G. K., **Chakraborty, C.**, and Das, P. K. *In search of an optimal public policy in a pandemic: The question of lives versus livelihood*. **Journal of Economic Analysis**, 2024.

Doctoral Research and Work in Progress

- **Reinforcement learning from human feedback (RLHF) and multi-turn RL:** Investigating learning from human feedback in multi-turn reinforcement learning settings with Prof. Cong Ma.
- **Nash learning from human feedback (NLHF):** Studying preference-based policy learning through game-theoretic formulations.
- **Learning from demonstrations:** Established matching sample-complexity bounds for standard and robust pass@k evaluation with imperfect demonstrators; developed a greedy multiplicative-weights learner.
- **Synthetic data and model collapse:** Studying how iterative training on synthetic data affects generative distributions and downstream classification, including conditions under which discriminative performance can persist.
- **Gaussian graphical models:** Working with Prof. Frederic Koehler on structure learning, aiming to develop polynomial-time algorithms with improved sample complexity.

Talks

- Presented research on generative modeling conducted with Prof. Larry Wasserman and Prof. Arun Kumar Kuchibhotla at the **University of Pennsylvania**.

Other Projects

- **Android development:** Developed basic Android applications.
- **Web scraping:** Completed Python web-scraping projects and built *YouScrap*, a command-line tool for retrieving YouTube video and channel metadata.
- **Arduino:** Completed projects using Arduino.
- **Cybersecurity:** Completed a course project for *Cyber Security and Ethical Hacking using Python*.

Courses, Camps, and Workshops

Fall 2020 Completed the 60-hour course **Cyber Security and Ethical Hacking using Python** at Ardent Collaborations.

Oct 2019 Attended an **AI and Machine Learning workshop** jointly organized by ISI and Intel.

Winter 2016 Attended the **INMO training camp** at North-Eastern Hill University, Shillong.

Winter 2015 Attended **Expedition**, a mathematics camp at G.C. College, Silchar, funded by the National Board for Higher Mathematics (NBHM).

Teaching Experience

- **Linear algebra tutor:** Currently tutor undergraduate students in linear algebra.
- **Programming instructor, PiParadox:** Taught R and Python over a three-month period.
- **Python workshop:** Led a three-day workshop delivered through YouTube Live.

Skills

Programming: Python, R, C, C++, SQL, Java, and shell scripting; basic PHP and Kotlin.

Document/Web: LaTeX and HTML; basic CSS.

Operating systems: Linux, macOS and Windows

Image editing: Basic GIMP.

Other Information

Languages: Bengali (native), English, Hindi, Odia, and Assamese; currently learning Sanskrit.

Hobbies and interests:

- Competitive programming.
- Chess (**FIDE-rated** player).
- Classical music; play the esraj and bansuri.
- Exploring Linux distributions.
- Reading.
- Bridge, including participation in local tournaments.